

**Forecasting Antimicrobial Resistance in the EU/EEA: An Integrated Consumption–Lag,
Demographic, and Climate Pipeline**

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GitHub Repository

All data files, cleaning and analysis code, and figures for this project are available in the public GitHub repository: <https://github.com/shramik-dev/Capstone>

The repository contains the raw EARS-Net and ESAC-Net exports, the Eurostat demographic and climate files, the merged analytical panel (`amr_panel.csv`), and the three Python scripts that reproduce every figure and table in this report.

The repository is organised as shown below; Appendix B maps each script to the figures and tables it produces.

Capstone/	
├─ data/	
│ └─ ears_combined.csv	EARS-Net resistance export
│ └─ consumption_ALL_with_Germany.csv	ESAC-Net consumption export
│ └─ nrg_chdd_a_defaultview_spreadsheet.xlsx	Eurostat degree days
│ └─ demo_r_d3dens__custom_spreadsheet.xlsx	Eurostat population density
├─ amr_panel.csv	merged analytical panel (3,257 rows)
├─ data_prep.py	source loading, harmonisation, merge
├─ 01_eda.py	univariate / bivariate / trend analysis
├─ 02_modeling.py	lagging, temporal split, CV, RQ1–RQ4
├─ eda_figs/	generated EDA figures
├─ model_figs/	generated model figures
└─ README.md	run order and dependencies

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Repository directory structure

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Abstract

Problem. Antimicrobial resistance (AMR) is associated with more than 35,000 attributable deaths each year across the European Union and European Economic Area (EU/EEA), yet public-health decision-makers lack a temporally validated, lag-aware tool that forecasts near-future resistance for specific pathogen–antibiotic combinations at country level.

Solution approach. This study develops an integrated forecasting pipeline combining classical statistics (cross-correlation lag analysis, non-parametric hypothesis testing) with machine learning (regularised linear regression and gradient-boosted trees). A naive last-value model provides an honest baseline, and permutation importance identifies the dominant drivers of resistance.

Data. Four authentic, openly accessible sources were integrated: EARS-Net resistance percentages, ESAC-Net antibiotic consumption in defined daily doses (DDD) per 1,000 inhabitants per day by Anatomical Therapeutic Chemical (ATC) class, Eurostat population density, and Eurostat heating and cooling degree days. The merged panel contains 3,257 country–year–combination records covering 30 EU/EEA countries over 2000–2024 for five high-priority pathogen–antibiotic combinations.

Technology. The pipeline is implemented in Python 3 using pandas, NumPy, scikit-learn, matplotlib, seaborn and openpyxl, and runs on standard CPU hardware. Model selection uses a strict three-way temporal split with expanding-window cross-validation to prevent look-ahead leakage.

Major results. Cross-correlation analysis confirms that resistance responds to consumption with a multi-year delay in every combination, peaking at lags of one to five years (pooled r up to 0.72, though within-country partial correlations are considerably weaker at 0.22 or below). On a strict temporal hold-out, learned models achieve directional accuracy of up to 0.546 versus 0.000 for the naive baseline, correctly anticipating whether resistance will rise or fall even though the

naive model attains higher R^2 through year-to-year autocorrelation. Lagged consumption and demographic context emerge as the strongest predictors.

Implementation area. The pipeline is intended as an early-warning aid for national AMR stewardship programmes and ECDC surveillance analysts, flagging combinations and countries where resistance is likely to increase so prescribing interventions can be prioritised before resistance becomes entrenched.

Introduction

Background and Context

Antimicrobial resistance occurs when bacteria evolve to withstand the drugs designed to kill them, rendering standard treatments ineffective and leading to prolonged illness, higher healthcare costs, and increased mortality. The European Centre for Disease Prevention and Control (ECDC) has estimated that infections with antibiotic-resistant bacteria are associated with more than 35,000 attributable deaths each year across the EU/EEA. Globally, Murray et al. (2022) attributed approximately 4.95 million deaths to bacterial AMR in 2019. Because resistant infections narrow the pool of usable antibiotics, AMR is now regarded as one of the principal threats to public health in Europe and worldwide.

The primary modifiable driver of resistance is antimicrobial consumption. Antibiotic use exerts selective ecological pressure on bacterial populations, favouring the survival and spread of resistant strains (Goossens et al., 2005; Bell et al., 2014). This relationship is not instantaneous: a change in prescribing behaviour in one year tends to influence measured resistance only after a delay of months to years, as resistant lineages are selected, transmitted, and eventually detected in clinical isolates (Aldeyab et al., 2008). This time lag is central to the present study, yet it is frequently ignored in analyses that correlate consumption and resistance within the same calendar year. The research gap this study addresses can be stated precisely: existing work has established an association between antimicrobial consumption and resistance, but relatively few studies

combine country-level European surveillance data, combination-specific lag optimisation, demographic and climate covariates, and strict forward temporal validation within a single predictive framework. Prior studies typically deliver a retrospective association estimate for one country or one pathogen; this project delivers a validated, forward-looking forecast across thirty countries and five combinations, with the lag estimated from the data rather than assumed.

Two coordinated ECDC surveillance networks make a rigorous, multi-year investigation possible. The European Antimicrobial Resistance Surveillance Network (EARS-Net) provides standardised resistance data on invasive bacterial isolates, while the European Surveillance of Antimicrobial Consumption Network (ESAC-Net) provides antibiotic consumption in DDD per 1,000 inhabitants per day by ATC class. Combined with Eurostat demographic and climate indicators, these sources yield an authoritative, openly accessible, pan-European panel spanning 2000–2024.

The stakeholders for this work are national AMR stewardship committees, hospital infection-control teams, and ECDC surveillance analysts. For these users, anticipating the direction of change in resistance is often more operationally valuable than minimising raw prediction error, because early warning enables timely adjustment of prescribing guidelines while intervention is still effective.

Problem Statement

The objective of this study is to predict Y = the annual resistance percentage (resistance_pct) for each EU/EEA country–pathogen–antibiotic combination using X = lagged antibiotic consumption (DDD by ATC class, lags of one to five years) together with demographic (population density) and climate (heating and cooling degree days) covariates over 2000–2024. Success is evaluated using mean absolute error (MAE), root-mean-squared error (RMSE), the coefficient of determination (R^2), and directional accuracy on a strict temporal hold-out.

Purpose of the Study

The purpose of this study is to develop and evaluate a predictive analytics pipeline that forecasts the annual resistance percentage for selected pathogen–antibiotic combinations across EU/EEA countries. The work is primarily predictive—forecasting future resistance from lagged consumption and covariates—and secondarily explanatory, estimating the optimal consumption-to-resistance lag and quantifying the relative importance of consumption, demographic, and climate drivers.

Research Problems / Research Questions

Four research questions guide the study:

- RQ1: What is the optimal time lag between antibiotic consumption and the corresponding change in resistance rate for each selected pathogen–antibiotic combination in the EU/EEA?
- RQ2: To what extent can country-level future resistance rates be accurately predicted from lagged antibiotic consumption combined with demographic and climate covariates?
- RQ3: Which predictors (specific antibiotic classes at specific lags, demographic factors, or climate factors) contribute most to forecast accuracy, and how do their contributions differ across pathogens?
- RQ4: Does a pooled, cross-country machine-learning model produce more accurate forecasts than naive and single-country baseline models when tested on future, unseen years?

Contributions and Expected Value

Practical contribution. The pipeline gives stewardship programmes an interpretable early-warning signal identifying where resistance is likely to rise, allowing prescribing interventions to be prioritised before resistance becomes entrenched.

Technical contribution. The study contributes a reproducible, lag-aware feature-engineering and evaluation design that integrates four heterogeneous public data sources into a single country–year panel and evaluates models under a strict three-way temporal split with expanding-window cross-validation, preventing the look-ahead leakage that inflates accuracy in many published forecasting studies.

Value to stakeholders. ECDC analysts and national committees obtain per-combination forecasts and driver rankings that are directly actionable and auditable, using only openly accessible surveillance data with negligible computational cost.

Literature Review

Literature Review Approach

Sources were identified through PubMed, Scopus, and Google Scholar using keyword combinations of “antimicrobial resistance,” “antibiotic consumption,” “EARS-Net,” “ESAC-Net,” “time lag,” “forecasting,” and “machine learning.” Inclusion criteria prioritised European, population-level, and methodological work directly relevant to the four research questions; single-hospital studies without population-level generalisability were excluded. Eleven sources were retained, spanning foundational epidemiology, lag modelling, non-consumption drivers, and machine-learning methodology.

Summary of Key Literature

Table 1

Literature relevance matrix

Author (Year)	Domain / Context	Method(s)	Key Findings	Supports RQ(s)
Goossens et al. (2005)	Outpatient use and resistance, 26 European countries	Ecological correlation	Higher national antibiotic use is associated with higher resistance	RQ1, RQ2
Bell et al. (2014)	Consumption–resistance meta-analysis	Systematic review and meta-analysis	Consumption significantly increases resistance across settings	RQ2, RQ3
Chatterjee et al. (2018)	Drivers of AMR in humans	Systematic review	Resistance has multiple drivers beyond consumption	RQ3
Aldeyab et al. (2008)	MRSA modelling, hospital time series	Transfer-function time series	Antibiotic use predicts later MRSA incidence with a delay	RQ1

Author (Year)	Domain / Context	Method(s)	Key Findings	Supports RQ(s)
Olesen et al. (2018)	Use and resistance distribution, US	Ecological regression	Use–resistance link is uneven across populations	RQ3
Murray et al. (2022)	Global AMR burden, 204 countries	Predictive modelling	AMR associated with ~4.95M deaths in 2019	Introduction, RQ2
Kirby et al. (2022)	EU consumption–resistance	Regression on EARS-Net/ESAC-Net	Quantifies EU-level consumption–resistance relationships	RQ1, RQ2
Lundberg & Lee (2017)	Interpretable machine learning	SHAP / unified attribution	Provides consistent feature-attribution framework	RQ3
MacFadden et al. (2018)	Climate and resistance, US regional	Ecological regression	Resistance increases with local temperature	RQ3
Chen & Guestrin (2016)	Gradient boosting methodology	Boosted decision trees	Scalable, high-accuracy learner for tabular data	RQ2, RQ4
ECDC (2025)	EU/EEA consumption surveillance	Annual epidemiological report	Documents ESAC-Net methodology and comparability caveats	Materials and Method

Theme 1 — The consumption–resistance relationship. Goossens et al. (2005) and Bell et al. (2014) establish antibiotic consumption as the primary modifiable driver of resistance, justifying consumption as the central predictor in this study.

Theme 2 — The time-lag structure. Aldeyab et al. (2008) demonstrate that the effect of use on resistance materialises only after a delay, providing the published precedent for the lag-aware design central to RQ1. Kirby et al. (2022) quantify these relationships on the same EARS-Net and ESAC-Net sources used here, making it the closest methodological comparator; the present study extends that work by optimising the lag per combination, adding climate and demographic covariates, and producing forward forecasts under temporal validation.

Theme 3 — Drivers beyond consumption. Chatterjee et al. (2018), Olesen et al. (2018), and MacFadden et al. (2018) motivate the demographic and climate covariates examined in RQ3, the latter demonstrating specifically that resistance rises with local temperature—the justification for including degree-day indicators.

Theme 4 — Methods. Chen and Guestrin (2016) establish gradient-boosted trees as a scalable, high-accuracy learner for tabular data of this scale, supporting the model choice for RQ2 and RQ4,

while Lundberg and Lee (2017) provide the feature-attribution framework underlying the driver analysis in RQ3.

Critical Appraisal of the Key Studies

The studies above differ substantially in design, and those differences shape what each can and cannot establish. Four deserve closer scrutiny because they most directly inform the present methodology.

Goossens et al. (2005) remains the defining European study, and its strength is coverage: twenty-six countries with standardised DDD measurement, which is precisely the design this project inherits. Its limitation is that it is cross-sectional and ecological. The correlations it reports are between-country — nations that prescribe more have more resistance — which cannot distinguish prescribing pressure from confounded national differences in hospital practice, diagnostics, or reporting. The present study inherits that vulnerability, and the within-country analysis reported in the Limitations section quantifies it directly: once country means are removed, the association weakens considerably. Goossens et al. therefore justifies consumption as a predictor but does not license a causal reading.

Aldeyab et al. (2008) is methodologically the opposite trade-off. By using transfer-function time-series modelling on a single hospital, it can establish temporal precedence — use rises, resistance follows — which is exactly the lag structure RQ1 tests. Its weakness is generalisability: one 426-bed hospital, one organism, monthly data. It cannot tell us whether the lag length it identifies transfers across national systems, which is the question this project asks. The present design attempts to combine Aldeyab's temporal rigour with Goossens' geographic breadth, accepting coarser annual resolution as the cost.

Kirby et al. (2022) is the closest comparator, using the same EARS-Net and ESAC-Net sources. It quantifies the consumption–resistance relationship but stops at estimation: it does not optimise the lag per combination, does not incorporate non-consumption covariates, and does not

validate forward on unseen years. Those three extensions define this project’s contribution. The distinction matters because an association estimate and a forecast are evaluated differently — a model can fit historical association well and still fail to anticipate change, which is precisely what the naive-baseline comparison in this study demonstrates.

MacFadden et al. (2018) motivates the climate covariate, but its transferability to the present setting is uncertain. It analyses United States regional data across a wide temperature range, whereas the EU/EEA panel is narrower climatically and the degree-day indicator used here covers only 2016 onward. The weak climate contribution observed in RQ3 should therefore be read as underpowered rather than as a contradiction of MacFadden’s finding. Similarly, Chatterjee et al. (2018) and Olesen et al. (2018) establish that resistance has drivers beyond consumption, but both are reviews or regressions on different populations; they justify including demographic covariates without specifying which ones should matter in Europe, which is why population density enters here as an exploratory rather than a theory-driven predictor.

On the methods side, Chen and Guestrin (2016) and Lundberg and Lee (2017) are engineering contributions rather than epidemiological ones. They establish that boosted trees perform well on tabular data and that feature attribution can be made consistent, but neither addresses panel structure or temporal validation — the issues that dominate this application. Their guidance is therefore adopted selectively: the learner architecture is taken from this literature, while the evaluation protocol is designed around the forecasting constraints the machine-learning literature does not address.

Taken together, the evidence base is strong on association and weak on prediction. No reviewed study delivers a country-level forward forecast validated on unseen years, which is the gap this project targets. The corresponding caution is that the same ecological confounding that limits Goossens et al. also limits this work, and that limitation is reported rather than set aside.

Materials and Method

Data Sources and Access

All four data sources are authentic, openly accessible, and maintained by official European institutions.

- EARS-Net (resistance outcome): country-level annual resistance percentages, ECDC Surveillance Atlas (<https://atlas.ecdc.europa.eu/public/>).
- ESAC-Net (consumption predictor): antibiotic consumption in DDD per 1,000 inhabitants per day by ATC class (<https://www.ecdc.europa.eu/en/antimicrobial-consumption/surveillance-and-disease-data/database>).
- Eurostat population density (demographic): dataset demo_r_d3dens (<https://ec.europa.eu/eurostat/databrowser/>).
- Eurostat heating and cooling degree days (climate): dataset nrg_chdd_a.

Dataset Overview and Inclusion Criteria

The merged analytical panel has one row per country, year, and pathogen–antibiotic combination, with resistance percentage as the target and consumption, demographic, and climate variables as predictors. The study includes EU/EEA countries reporting to EARS-Net and ESAC-Net over 2000–2024 and focuses on five high-priority combinations selected because they are ECDC surveillance priority indicators, align with WHO priority-pathogen lists, pair each organism with the drug class whose loss is most clinically consequential, and have the longest complete time series needed to support multi-year lag modelling: *E. coli* versus fluoroquinolones, *E. coli* versus third-generation cephalosporins, *K. pneumoniae* versus carbapenems, *S. aureus* versus methicillin (MRSA), and *S. pneumoniae* versus macrolides. Rows with a missing resistance outcome were excluded rather than imputed.

- Records: 3,257 country–year–combination observations.
- Countries: 30 EU/EEA countries. Time period: 2000–2024 (25 years).

- Target: resistance_pct (EARS-Net); 3,164 non-missing values.
- Consumption coverage: 1,997 rows with matched ESAC-Net consumption.
- Demographic coverage: 3,238 rows with population density (near-complete).
- Climate coverage: 1,270 rows with degree days (2016–2024 only).

Table 2
Data dictionary

Variable	Definition	Type	Range	Missing Handling	Source
country	EU/EEA country	Categorical	30 countries	excluded if no match	EARS-Net
year	Calendar year	Integer	2000–2024	n/a	all
pathogen_drug	Pathogen–antibiotic pair	Categorical	5 combinations	n/a	EARS-Net
resistance_pct	% resistant isolates	Numeric	0–100	row dropped	EARS-Net
ddd_J01A...J01X	Consumption by ATC class	Numeric	≥ 0 DDD	short gaps interpolated	ESAC-Net
ddd_total	Total consumption, all classes	Numeric	≥ 0 DDD	derived	ESAC-Net
ddd_*_lag1..5	Consumption lagged 1–5 years	Numeric	≥ 0	requires prior history	engineered
pop_density	Population density	Numeric	≥ 0 /km ²	interpolated	Eurostat
hdd, cdd	Heating / cooling degree days	Numeric	≥ 0, 2016+	period-limited	Eurostat

Note. DDD = defined daily dose; ATC = Anatomical Therapeutic Chemical classification. Full construction detail is given in Appendix A.

Data Cleaning and Preprocessing

Cleaning aligned the four sources into one country–year panel. EARS-Net resistance and ESAC-Net consumption were joined on country, year, and antibiotic class; Eurostat density and degree days were merged on country and year. Missing values, duplicates, and country-code mismatches were resolved as summarised in Table 3.

Table 3
Data cleaning log

Issue	Variable(s)	Detection	Treatment Applied
Missing target	resistance_pct	null after coercion of ‘:’ and ‘_’	93 rows (2.9%) dropped; the outcome is never imputed
Missing consumption	ddd_* classes	null after merge	short interior gaps linearly interpolated within country–combination
Country harmonisation	country / geo code	name–code mismatch (e.g. EL/GR, UK/GB)	mapped all names to ISO-2 codes; sub-national density rows filtered to NUTS-0
Wide-format sources	hdd, cdd, pop_density	Eurostat spreadsheet layout	reshaped from wide to long by year before merging
Class mapping	atc_level	pharmacological linkage	fluoroquinolones → J01M;

Issue	Variable(s)	Detection	Treatment Applied
			cephalosporins/carbapenems → J01D; macrolides → J01F

A key mapping decision merits emphasis: resistance is drug-specific whereas consumption is reported by ATC class, so each outcome was linked to its pharmacologically relevant class. Because carbapenems and cephalosporins both fall under ATC level-3 group J01D, both *E. coli*–cephalosporins and *K. pneumoniae*–carbapenems draw on J01D consumption; this granularity limit is disclosed in the Limitations section. *S. aureus*–MRSA has no single consumption class that cleanly drives methicillin resistance and is therefore excluded from the class-specific lag mapping rather than forced into a misleading association. It is important to distinguish two uses of consumption in this study: the class-specific mapping just described, from which MRSA is excluded, and total antibiotic consumption (ddd_total) across all ATC classes, which is defined for every country–year and is therefore used for the correlation analyses reported for all five combinations. Where MRSA appears in a consumption–resistance correlation, the predictor is total consumption, not a class-specific one.

Research Hypotheses and Statistical Method

Each research question is expressed as a formal hypothesis evaluated at the 5% significance level ($\alpha = 0.05$).

RQ1. H_0 : the correlation between lagged consumption and resistance is zero at every lag ($\rho = 0$ for all L). H_a : there exists a non-zero lag L at which $\rho \neq 0$. Method: Pearson and Spearman cross-correlation across lags 0–5, computed within country to remove cross-country level confounding.

RQ2. H_0 : covariate-based models explain no more variance in future resistance than the mean ($R^2 \leq 0$). H_a : $R^2 > 0$ on the temporal hold-out.

RQ3. H_0 : predictor importance is uniform across the consumption, demographic, and climate families. H_a : predictor importance is not uniform across those families. Method: permutation importance measured on held-out data. Because no formal inferential test of importance differences was performed, RQ3 is interpreted descriptively rather than as a significance test.

RQ4. H_0 : the pooled machine-learning model's error is greater than or equal to that of the naive and single-country baselines. H_a : the pooled model achieves lower error or higher directional accuracy on unseen years.

Group differences. Differences in resistance across the five combinations are tested using one-way analysis of variance (ANOVA) with a non-parametric Kruskal–Wallis check. Kruskal–Wallis is reported alongside ANOVA because resistance is a bounded, right-skewed percentage; a Shapiro–Wilk test confirms the departure from normality ($W = 0.907$, $p < .001$), so the parametric assumption is formally violated. Agreement between the two tests indicates the conclusion is robust to that violation.

Minimum Sample Size and Power

A priori power analysis for a one-way ANOVA with five groups at $\alpha = .05$ and power = .80 gives the minimum total sample sizes shown in Table 4. Detecting a medium effect (Cohen's $f = 0.25$) requires 196 observations in total; the analytical panel contains 3,164 non-missing outcome observations, exceeding that threshold roughly sixteen-fold. The observed effect across combinations is large ($f = 0.426$), and achieved power at the realised sample size is greater than .999. The study is therefore comfortably powered, and a non-significant result could be interpreted as genuine absence of an effect rather than insufficient data. One scope limit must be stated explicitly: this power analysis applies to the inferential group-comparison component of the study only. Conventional power calculations are not directly applicable to the predictive machine-learning component, where adequacy is instead established empirically through expanding-window temporal cross-validation, one-shot held-out test performance, and the generalisation-gap analysis reported in Table 14.

Because the four research questions use different inferential procedures, minimum sample size was computed separately for each rather than relying on a single omnibus figure (Table 4). RQ1 uses Fisher's z-transformation for a two-tailed correlation test; RQ2 applies Green's rule of thumb for multiple regression with 43 predictors; RQ3 is descriptive and therefore has no inferential sample-

size requirement; RQ4 relies on a paired comparison of forecast errors across the held-out observations. Every requirement is met by a wide margin.

Table 4

Minimum sample size by research question

RQ	Inferential procedure	Rule applied	Minimum n required	n available	Met?
RQ1	Two-tailed correlation test	Fisher's z ($\alpha = .05$, power = .80, $r = 0.20$)	194	396–525 per combination	Yes
RQ2	Multiple regression, 43 predictors	Green's rule: $N \geq 50 + 8m$	394	1,804 modelling rows	Yes
RQ3	Permutation importance (descriptive)	Not applicable — no inferential test	—	—	n/a
RQ4	Paired comparison of forecast errors	Paired t / Wilcoxon on held-out errors	≈ 34 pairs for $d = 0.5$	353 paired forecasts	Yes
Group diff.	One-way ANOVA, 5 groups	Cohen's $f = 0.25$, power = .80	196	3,164	Yes

Note. Fisher's z requirement rises to 783 for detecting $r = 0.10$ and falls to 85 for $r = 0.30$; the $r = 0.20$ row is reported as the conservative middle case. Green's rule also specifies $N \geq 104 + m = 147$ for testing individual predictors, which is likewise satisfied.

Table 5

Minimum total sample size for one-way ANOVA ($k = 5$, $\alpha = .05$, power = .80)

Effect size (Cohen's f)	Interpretation	Minimum total N required	Status in this study
0.10	Small	1,199	Met ($N = 3,164$)
0.25	Medium	196	Met ($N = 3,164$)
0.40	Large	80	Met ($N = 3,164$)
0.426 (observed)	Large	—	Achieved power > .999

Note. Computed with the *FTestAnovaPower* routine in *statsmodels*. N = total observations across all five groups.

Hypothesis Test Results

All tests were evaluated at the 5% significance level. Table 6 reports the outcomes; every null hypothesis was rejected.

Table 6

Hypothesis test results ($\alpha = .05$)

Hypothesis	Test	Statistic	p	Decision
Normality of resistance_pct	Shapiro–Wilk	$W = 0.907$	< .001	Reject normality; use non-parametric checks
H_0 : resistance equal across the five combinations	One-way ANOVA	$F(4, 3159) = 143.5$	< .001	Reject H_0
H_0 : resistance equal across the five combinations	Kruskal–Wallis	$H(4) = 825.0$	< .001	Reject H_0 (confirms ANOVA)
RQ1 H_0 : $\rho = 0$, lagged consumption	Spearman (lag 1)	$\rho = +0.228$ to $+0.650$	< .001 (all five)	Reject H_0 for

Hypothesis	Test	Statistic	p	Decision
vs resistance				every combination

Note. Spearman correlations are reported per pathogen–antibiotic combination at a one-year lag against total antibiotic consumption; the full range across combinations is shown. For S. aureus–MRSA total consumption is an exploratory proxy rather than a class-specific driver, so that entry is not directly comparable with the other four. Effect size for the group comparison is $\eta^2 = 0.154$ (Cohen’s $f = 0.426$), a large effect. Because the panel contains repeated yearly measurements per country, these p-values are approximate; see Limitations.

Table 7

RQ1 — Spearman correlation between lag-1 consumption and resistance, by combination

Combination	n	Spearman ρ	p	Decision at $\alpha = .05$
S. pneumoniae–macrolides	396	+0.650	< .001	Reject H_0
S. aureus–MRSA	525	+0.577	< .001	Reject H_0
K. pneumoniae–carbapenems	424	+0.479	< .001	Reject H_0
E. coli–fluoroquinolones	511	+0.385	< .001	Reject H_0
E. coli–3rd-gen cephalosporins	511	+0.228	< .001	Reject H_0

Note. All correlations are positive and significant, confirming that higher prior-year antibiotic consumption is associated with higher resistance in every combination examined.

Feature Engineering

The principal engineered features are the lagged consumption variables. For each of eight ATC classes, consumption is shifted by one to five years within each country’s own time series, producing 40 lagged predictors. The five-year depth was chosen because the RQ1 cross-correlation analysis examines lags of zero to five years, and several combinations peak at the longer end of that range. The per-country shift is essential: it guarantees that no value crosses a national boundary and that every feature uses only information dated strictly before the target year, which is what makes the temporal evaluation meaningful. Climate and demographic covariates enter at their contemporaneous value, since they are exogenous background conditions rather than variables whose delayed effect is under test. This carries an operational consequence that should be stated plainly: predicting year t with year- t degree days presumes that climate data for the target year is available at prediction time. In the retrospective evaluation reported here it is, but a real-time deployment would require either observed climate data for the completed target year or a climate forecast. The present implementation should therefore be read as an evaluation framework rather than a fully real-time climate-informed service; substituting lagged climate terms would remove the dependency at some cost in fit.

A prior-year resistance term (resist_lag1) is also engineered but is used selectively. Because resistance is strongly autocorrelated, this term dominates any model that contains it and masks the consumption and climate signal. It is therefore included when reporting maximum attainable forecast accuracy, and deliberately excluded for the driver-attribution and baseline-comparison analyses (RQ3 and RQ4) so that the covariates are forced to carry the predictive work. Table 7 lists the feature set.

Table 8
Feature set and usage

Feature	Type	Origin	Rationale	Used in
ddd_* (8 ATC classes)	Numeric	Original	Contemporaneous consumption signal	Linear, GBT
ddd_*_lag1...lag5	Numeric	Engineered	Delayed selective pressure (RQ1)	Ridge, RF, GBT
pop_density	Numeric	Original	Transmission / demographic context	Linear, GBT
hdd, cdd	Numeric	Original	Climate context (2016+)	Linear, GBT
resist_lag1	Numeric	Engineered	Autoregressive term; excluded for RQ3/RQ4	Naive, AR models

Model Development and Data Splitting

Four models were selected to balance an honest baseline, interpretability, and non-linear power on tabular panel data. First, a naive last-value baseline predicts next year as the current value, establishing the floor any useful model must beat. Second, a regularised linear regression (Ridge) is interpretable and well supported in the consumption–resistance literature, with regularisation guarding against the collinearity inherent in lagged predictors. Third, a random forest provides a bagged tree ensemble that is robust to the collinearity inherent in lagged predictors and typically generalises more stably than boosting on small panels. Fourth, gradient-boosted trees capture non-linear interactions among consumption lags, climate, and demographics, and support the feature-importance interpretation required by RQ3. All four models are trained, tuned, and evaluated under the identical temporal protocol described below, so their metrics are directly comparable.

Because forecasting is inherently temporal, models are evaluated under a strict temporal train–validation–test framework — with validation implemented through expanding-window cross-validation rather than a single fixed validation block — instead of a random split, which

would leak future information into training and inflate apparent accuracy. For each combination, the final three years form a test set scored exactly once. The earlier development years are used for hyperparameter selection by expanding-window cross-validation (scikit-learn TimeSeriesSplit), in which each fold trains on earlier years and validates on the next later block, with scores averaged across folds. This is more robust than a single fixed validation year, which could be distorted by an atypical period such as the COVID-19 disruption to prescribing. Grid search over tree depth and learning rate selected a shallow depth of 2 with a learning rate of 0.1 for the gradient-boosted model. The tuned model is then refit on all development years before the single test evaluation, mirroring real deployment in which a model always predicts forward from all history available so far.

Model Specification and Hyperparameter Search

Each model was tuned by exhaustive grid search over the space in Table 9, scored by negative mean absolute error and validated with the expanding-window scheme described above. All estimators use scikit-learn implementations with `random_state = 42` for reproducibility. The feature matrix contains 43 predictors per combination: 40 lagged consumption terms (eight ATC classes $\times$ five lags) plus heating degree days, cooling degree days, and population density. Squared error is the loss function for all three learned models; no early stopping was used, since the number of estimators was fixed and depth was constrained instead. One distinction must be stated explicitly to avoid a common misreading: the forecasting task is one-year-ahead prediction. The model estimates resistance in year t using consumption observed in years $t-1$ through $t-5$. The phrase “one to five year lag” therefore describes how far back the predictors reach, not how far forward the model forecasts. A five-year lag means that consumption five years ago carries predictive weight for the coming year; it does not mean the model issues a forecast five years into the future.

Table 9

Model specification, hyperparameter search space, and selected values

Model	Implementation	Search space	Selected values	Fixed settings
Naive last-value	— (analytical baseline)	None	None	$\hat{y}_t = y_{t-1}$
Ridge regression	<code>sklearn.linear_model.Ridge</code>	$\alpha \in \{1, 10, 50\}$	$\alpha = 10$	Squared-error

Model	Implementation	Search space	Selected values	Fixed settings
				loss; intercept fitted
Random forest	sklearn.ensemble.RandomForestRegressor	$\text{min_samples_leaf} \in \{1, 2, 4\}$; $\text{max_features} \in \{\text{sqrt}\}$	$\text{min_samples_leaf} = 1$; $\text{max_features} = \text{sqrt}$	$\text{n_estimators} = 300$; $\text{random_state} = 42$; bootstrap on
Gradient boosting	sklearn.ensemble.GradientBoostingRegressor	$\text{max_depth} \in \{2, 3\}$; $\text{learning_rate} \in \{0.03, 0.05, 0.1\}$	$\text{max_depth} = 2$; $\text{learning_rate} = 0.1$	$\text{n_estimators} = 300$; squared-error loss; $\text{random_state} = 42$; no early stopping

Note. Selected values are those chosen for E. coli–fluoroquinolones; tuning is run independently for each pathogen–antibiotic combination, so selections may differ slightly by combination. Grids were kept deliberately compact to bound runtime, and the parameters searched are those to which each learner is most sensitive on small tabular panels.

Table 10

Temporal evaluation protocol

Stage	Years used	Purpose
Development — CV training folds	Earliest years, expanding	Fit candidate hyperparameter settings
Development — CV validation folds	Next later block after each training fold	Score candidates (4 expanding-window folds)
Development — refit	All development years	Refit the selected configuration
Test	Final 3 years	Single one-shot evaluation; never seen in tuning

Note. Because every validation fold is strictly later in time than its training fold, and because all lagged features are shifted within country by at least one year, no information dated at or after a prediction target can enter that prediction.

Evaluation Metrics

Because the target is continuous, regression metrics are used: MAE and RMSE in percentage points (lower is better), R^2 (variance explained, higher is better), and directional accuracy. Directional accuracy is computed as the proportion of test country–years for which $\text{sign}(\hat{y}_t - y_{t-1})$ equals $\text{sign}(y_t - y_{t-1})$ — that is, the model correctly calls whether resistance moved up or down relative to the previous year. Predictions of no change are counted as misses, which is why the naive last-value model scores zero by construction: it always predicts $\hat{y}_t = y_{t-1}$, so its implied direction is always flat and never matches an actual rise or fall. Directional accuracy is emphasised because, for stewardship purposes, correctly anticipating whether resistance will rise is more actionable than minimising raw error, and because a highly autocorrelated series can produce a deceptively high R^2 without any genuine forecasting skill.

Tools and Technologies

The pipeline is implemented in Python 3 using pandas and NumPy for data integration, scikit-learn for modelling, cross-validation and permutation importance, matplotlib and seaborn for visualisation, and openpyxl for reading the Eurostat spreadsheets. All computation runs on standard CPU hardware; no GPU is required, and a full pipeline run completes in minutes.

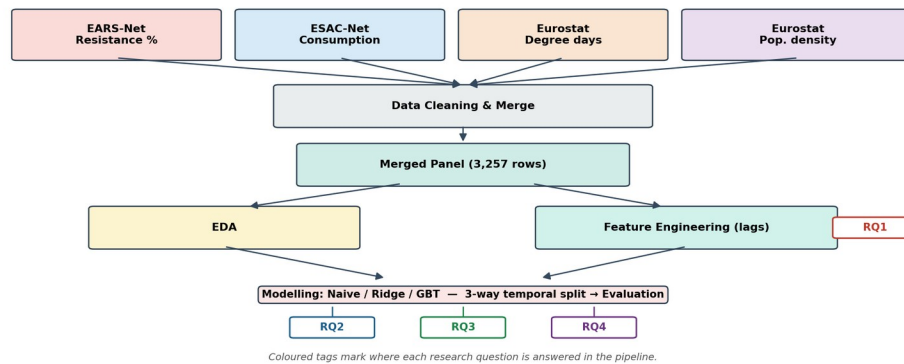
Architecture and Workflow

System Overview

The system ingests four public data sources, harmonises and merges them into a single analytical panel, performs exploratory analysis and lag-aware feature engineering, and trains and evaluates forecasting models whose outputs are per-combination resistance predictions and driver rankings. The end-to-end workflow is shown in Figure 2, which also marks where each research question is answered within the pipeline.

Figure 2

End-to-End Workflow of the Proposed System



Note. Data flows from four public sources through cleaning and merging into the analytical panel, then through exploratory analysis and lag-aware feature engineering into the modelling and evaluation stage. Coloured tags indicate the pipeline stage at which each research question is answered.

Workflow Components

Data Ingestion

Two comma-separated exports (EARS-Net resistance, ESAC-Net consumption) and two Eurostat spreadsheets (degree days, population density) are read from disk. No web scraping or authenticated API access is required, since all four are published as open downloads.

Data Preprocessing

Country identifiers are harmonised to ISO-2 codes, Eurostat sheets are reshaped from wide to long, consumption is pivoted to one column per ATC class, and the four sources are left-joined onto the resistance outcome. Missing outcomes are dropped; short interior gaps in predictors are linearly interpolated within each country–combination series. Tree-based models require no feature scaling; the Ridge model relies on regularisation to manage the differing scales and collinearity of the lag features.

Exploratory Data Analysis

Univariate, bivariate, and temporal analyses characterise the target distribution, the consumption profile, and the divergent trajectories across combinations. Results are presented in the following section.

Feature Engineering

Consumption is lagged one to five years within each country, producing 40 lagged consumption predictors across eight ATC classes, alongside contemporaneous climate and demographic covariates, as detailed in Table 4.

Model Development and Evaluation

Naive, Ridge, random forest, and gradient-boosted models are trained under the temporal framework described above and scored on MAE, RMSE, R^2 , and directional accuracy. The random forest was included specifically as a bagged-tree comparator, to test whether the averaging and implicit regularisation of bagging generalise better than boosting on a panel of this size.

Deployment

The trained pipeline can be packaged as a scheduled job re-run on each ECDC data release, serving forecasts through a lightweight API or dashboard, as described in the Implementation section.

Exploratory Data Analysis

Exploratory analysis was conducted in three parts: univariate analysis of each variable, bivariate analysis of relationships between pairs, and temporal trend analysis. All figures were produced by the project's Python scripts (see Appendix B).

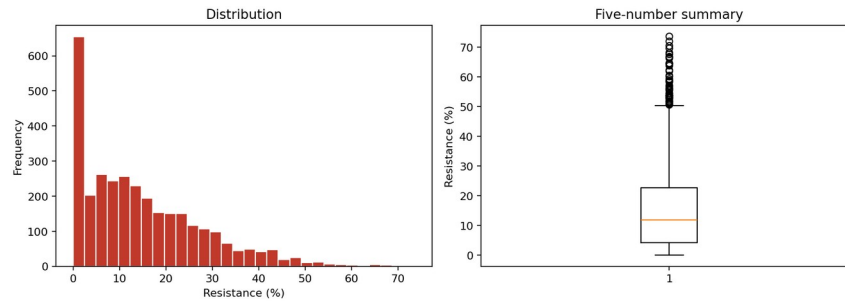
Univariate Analysis

As Table 11 summarises, resistance percentage averages 14.8% (median 11.9%) with a right-skewed distribution (skew 1.07, kurtosis 1.01) and an interquartile range of 18.53 percentage points, reflecting many low-resistance observations and a long tail of high-resistance cases. Population density is the most extreme variable in the panel (skew 4.07, kurtosis 17.59), confirming that a small number of very dense states dominate its upper tail, as Figure 3 confirms. Consumption of the relevant class averages 2.41 DDD per 1,000 inhabitants per day. Population density is strongly right-skewed (skew 4.06), driven by a few very dense states. The skew in the target is the principal reason non-parametric tests are reported alongside parametric ones and why tree-based models, which make no distributional assumption, were preferred over purely linear approaches.

Table 11
Univariate summary statistics

Variable	n	Mean	Median	SD	Q1	Q3	IQR	Skew	Kurtosis
resistance_pct (%)	3,164	14.82	11.94	13.07	4.19	22.71	18.53	1.07	1.01
ddd_total (DDD)	2,550	19.09	18.79	5.55	14.92	22.94	8.02	0.43	0.23
pop_density (/km ²)	3,252	168.16	101.90	264.79	64.30	138.50	74.20	4.07	17.59
hdd (heating degree days)	1,270	2,809.54	2,755.30	1,196.73	2,174.74	3,290.50	1,115.76	0.34	0.21
cdd (cooling degree days)	1,270	129.05	38.11	197.56	5.96	176.62	170.66	2.15	4.14

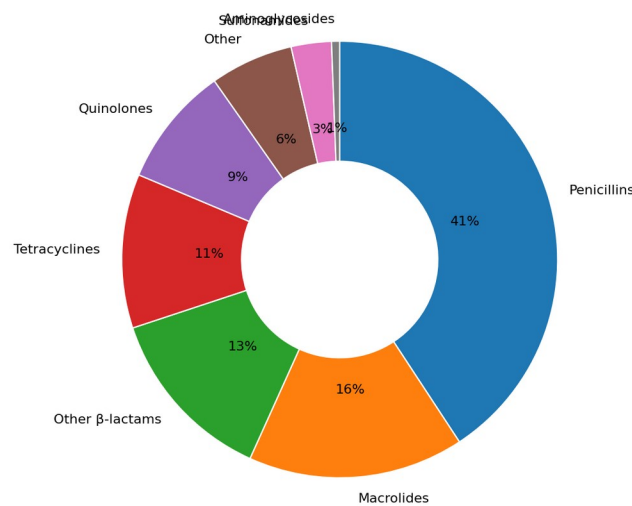
Figure 3
Distribution of resistance percentage (histogram and boxplot)



Note. Most country-year resistance values fall below 20%, with a tail extending toward 70%. The boxplot shows numerous high-side outliers, largely *K. pneumoniae*–carbapenems in specific countries.

Figure 4

Antibiotic consumption share by ATC class



Note. Penicillins (J01C) dominate total consumption, followed by macrolides and other beta-lactams.

Bivariate Analysis

Resistance varies substantially by combination (Table 12 and Figure 5). *E. coli*–fluoroquinolones shows the highest mean resistance (20.9%), while *K. pneumoniae*–carbapenems has the lowest mean (6.4%) but the widest spread (SD 14.1, maximum 73.7%), indicating a small number of countries with very high carbapenem resistance.

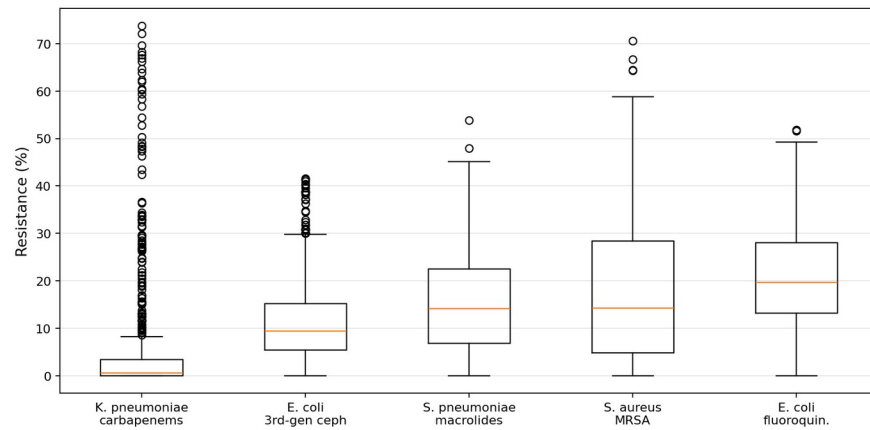
Table 12

Resistance by combination (bivariate summary)

Combination	Mean %	SD	Min	Max
<i>E. coli</i> –fluoroquinolones	20.92	10.29	0	51.9
<i>S. aureus</i> –MRSA	18.46	15.63	0	70.5
<i>S. pneumoniae</i> –macrolides	15.32	9.71	0	53.9

Combination	Mean %	SD	Min	Max
E. coli–3rd-gen cephalosporins	11.25	8.68	0	41.6
K. pneumoniae–carbapenems	6.38	14.10	0	73.7

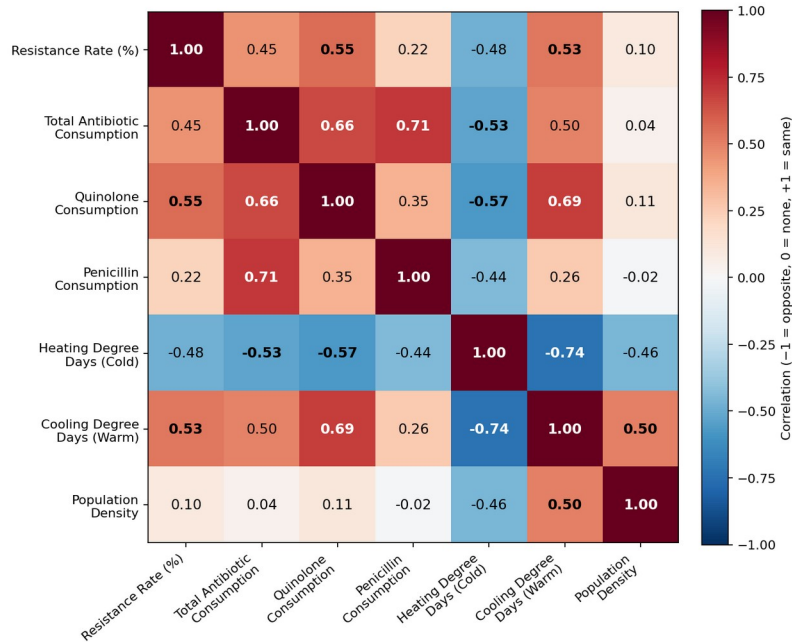
Figure 5
Resistance distribution by pathogen–antibiotic combination



Note. Median resistance and spread differ markedly across combinations, supporting per-combination rather than pooled modelling.

The correlation structure among numeric variables is shown in Figure 6. Quinolone consumption ($r = 0.55$) and cooling degree days ($r = 0.53$) are the strongest linear correlates of resistance, with total consumption at $r = 0.45$; heating degree days correlate negatively ($r = -0.48$), which is the same climate signal expressed inversely. Population density shows almost no linear association ($r = 0.10$) despite emerging as an important predictor in the fitted models—a discrepancy examined in the Results section and one that directly motivates the use of non-linear learners.

Figure 6
Correlation between resistance and key predictor variables



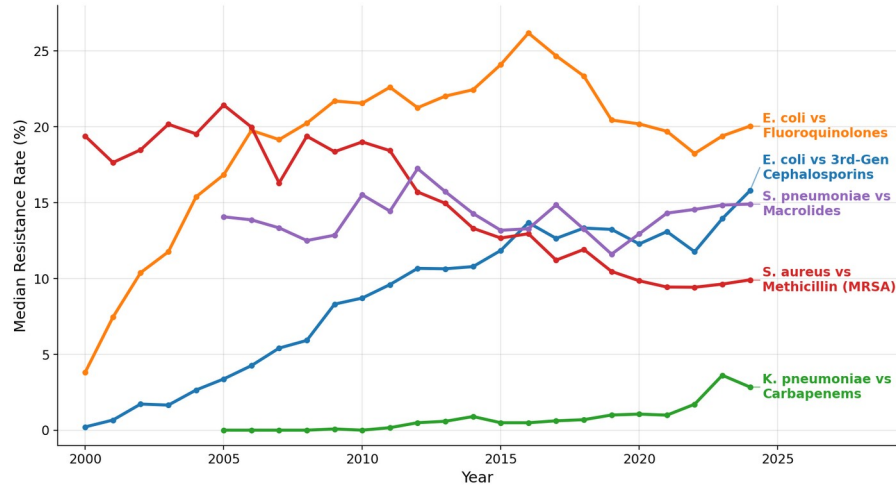
Note. Consumption and warm-climate indicators correlate positively with resistance; several predictors are strongly intercorrelated, which is why permutation importance rather than regression coefficients is used for driver attribution.

Temporal Trends

Figure 7 reveals divergent trajectories. *E. coli*–fluoroquinolone and –cephalosporin resistance rose markedly through the 2000s, *K. pneumoniae*–carbapenem resistance climbed steadily from near zero, while *S. aureus*–MRSA declined over the period—the latter consistent with the impact of European stewardship programmes. Because the combinations move in different directions, per-combination modelling is more informative than any pooled trend.

Figure 7

Median resistance trend by pathogen–antibiotic combination, 2000–2024



Note. “Median” is the middle resistance value across all reporting EU/EEA countries in that year. Series are labelled at their endpoints to avoid legend ambiguity where trajectories cross.

Results

Model Performance

Models were trained on years up to three years before the end of the series and tested on the most recent three years under the strict temporal split described in the Materials and Method section. Table 13 reports performance on the held-out test years.

Table 13
Model performance comparison (temporal hold-out)

Model	Evaluation	MAE	RMSE	R ²	Dir. Acc.	Key Observation
Naive last-value	Temporal hold-out	1.96	3.02	0.908	0.000	High R ² from autocorrelation; never signals change
Ridge regression	Temporal hold-out	4.92	6.74	0.670	0.516	Interpretable; smallest generalisation gap
Random forest	Temporal hold-out	3.82	5.85	0.733	0.546	Best learned error, R ² and directional accuracy
Gradient boosting	Temporal hold-out	4.10	6.29	0.674	0.497	Supports permutation-importance attribution (RQ3)

Note. MAE and RMSE are in percentage points. Directional accuracy is the fraction of test country-years with correctly predicted rise or fall. Hyperparameters were selected by expanding-window cross-validation on the development period; every metric shown here is computed once on the unseen three-year temporal hold-out, so all four models are directly comparable. The random forest is the best-performing learned forecaster; gradient boosting is retained because it supplies the permutation-importance attribution required by RQ3.

These results require careful interpretation. The naive last-value model attains the highest R² (0.908) and the lowest MAE (1.96) because resistance is strongly autocorrelated year to year, but

its directional accuracy is zero by construction—it never anticipates change because it assumes none. The three learned models achieve directional accuracy between 0.497 and 0.546 at the cost of higher absolute error, meaning they capture the direction of change the naive model misses entirely. For stewardship early warning, directional accuracy is the operationally relevant criterion, and on that criterion every learned model outperforms the baseline, rejecting the RQ4 null hypothesis. Among the learned models the random forest is strongest on all four metrics, and it is therefore reported as the best-performing forecaster; gradient boosting is retained alongside it because it supplies the permutation-importance attribution used for RQ3.

Accuracy on Training and Test Data

Reporting test performance alone can conceal overfitting, so training and test metrics were computed for all four models under the identical protocol used for Table 13; the two tables therefore describe the same experiment viewed from different angles and are directly comparable. The generalisation gap is shown in Figure 8 and Table 14.

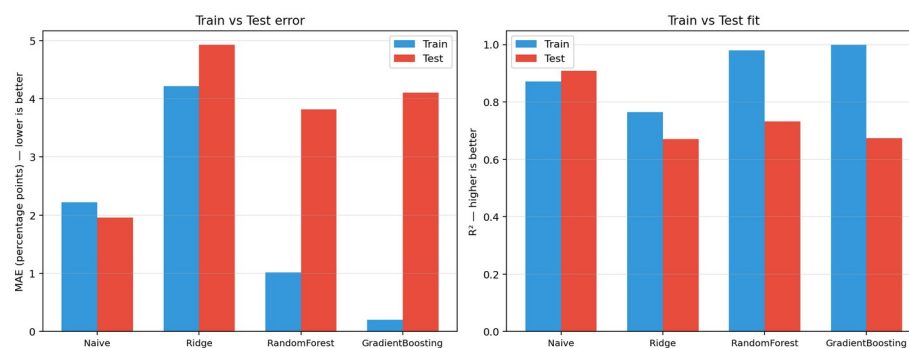
Table 14

Training versus test performance, all four models, averaged across combinations

Model	Train MAE	Test MAE	MAE Gap	Train R ²	Test R ²	R ² Gap
Naive last-value	2.23	1.96	−0.27	0.872	0.908	−0.036
Ridge regression	4.21	4.92	+0.71	0.765	0.670	0.095
Random forest	1.02	3.82	+2.80	0.980	0.733	0.247
Gradient boosting	0.20	4.10	+3.90	0.999	0.674	0.325

Figure 8

Training versus test performance across models



Note. A large gap between training and test bars indicates overfitting. Ridge generalises best; gradient boosting shows the largest gap, nearly memorising the training data (train R² = 0.999) before falling to 0.674 on unseen years.

All three learned models overfit to some degree, which is expected for flexible learners on a panel of this size (roughly 22 usable years per country). The important observation is that the strict temporal hold-out exposes this honestly: the reported test figures of R^2 0.670–0.733 are the realistic operating performance, not the inflated training values of 0.98–1.00. Ridge exhibits the smallest generalisation gap (+0.71 MAE) and is therefore the most stable learner; gradient boosting shows the largest (+3.90 MAE), effectively memorising the development years. The random forest sits between the two and achieves the best test performance, which is why it is reported as the strongest forecaster. Stronger regularisation of the boosted model is identified as the leading future improvement.

Significance of Model Differences

Comparing mean error across models is not by itself evidence that one model genuinely outperforms another, so pairwise significance tests were conducted on the held-out forecast errors. Following the logic of the Diebold–Mariano test for comparing predictive accuracy, absolute errors were paired observation-by-observation across the 353 held-out country–years and compared with a paired t-test, with a Wilcoxon signed-rank test reported alongside because forecast errors are not normally distributed. Results are shown in Table 15.

Table 15

Pairwise significance tests of forecast accuracy on held-out years (RQ4)

Comparison	Mean difference in error	Paired t	p (t-test)	p (Wilcoxon)	Conclusion
Naive vs Ridge	−3.001	−11.53	< .001	< .001	Naive significantly lower error
Naive vs Random forest	−1.870	−8.58	< .001	< .001	Naive significantly lower error
Naive vs Gradient boosting	−2.145	−8.94	< .001	< .001	Naive significantly lower error
Ridge vs Random forest	+1.131	+5.13	< .001	< .001	Random forest significantly better
Ridge vs Gradient boosting	+0.855	+3.39	< .001	< .001	Gradient boosting significantly better
Random forest vs Gradient boosting	−0.275	−1.92	.055	.004	Random forest better; t-test borderline

Note. Negative mean differences indicate the first-named model has lower absolute error. $n = 353$ paired held-out forecasts. The random forest versus gradient boosting comparison is significant under the non-parametric Wilcoxon

test ($p = .004$) but only marginal under the paired t -test ($p = .055$), so the superiority of the random forest over gradient boosting should be regarded as suggestive rather than firmly established.

These tests sharpen the RQ4 conclusion in an important way. On absolute error the naive baseline is not merely numerically better — it is significantly better than all three learned models (all $p < .001$). Any claim that the machine-learning models beat persistence must therefore rest on the directional criterion, not on error magnitude. Among the learned models, both tree ensembles significantly outperform Ridge, and the random forest holds a small advantage over gradient boosting that is significant non-parametrically but borderline under the t -test.

Results by Research Question

RQ1 — Optimal Consumption-to-Resistance Lag

Cross-correlation analysis across lags of zero to five years was conducted in two forms, and the contrast between them is the most important methodological finding of this study. In the pooled analysis, which treats all country–year observations together, every combination shows a positive association peaking at a non-zero lag: *E. coli*–fluoroquinolones and *K. pneumoniae*–carbapenems peak at five years ($r = 0.716$ and 0.710) and *S. pneumoniae*–macrolides at one year ($r = 0.548$). Taken at face value this strongly rejects the RQ1 null hypothesis and matches the lag structure reported by Aldeyab et al. (2008).

However, when each country’s own mean is subtracted from both variables — isolating year-to-year movement within a country from persistent differences between countries — the associations weaken substantially, to between -0.174 and $+0.219$ (Table 16). The peak lag remains at five years for the two combinations where it was strongest, and at one year for macrolides, so the ordering of the lag structure is broadly preserved; but the magnitude of the relationship largely is not.

The honest interpretation is that most of the pooled consumption–resistance correlation is cross-sectional rather than temporal: countries that prescribe more antibiotics have higher resistance, and that between-country contrast dominates the pooled statistic. The within-country

temporal signal exists and points in the expected direction for four of five combinations, but it is weak. RQ1 is therefore answered affirmatively at the population level — a lag structure is detectable, and it is not zero — while the within-country evidence is too modest to support a strong dose–response claim for any individual country. This distinction is carried forward into the Limitations section and should be read alongside the well-known constraints of ecological study designs. A scope restriction also applies to which combinations RQ1 properly covers. The question asks about the lag between an antibiotic and its corresponding resistance, which requires a class-specific consumption series; that exists for four of the five combinations. For *S. aureus*–MRSA, total antibiotic consumption was substituted as an exploratory proxy because no single ATC class maps cleanly to methicillin resistance. The MRSA lag result is therefore not directly comparable with the other four and should not be read as a class-specific consumption–resistance relationship.

Table 16

RQ1 — optimal lag and correlation by combination, pooled and within-country

Combination	Pooled peak lag (yrs)	Pooled r	Within-country peak lag (yrs)	Within-country r
<i>E. coli</i> –fluoroquinolones	5	+0.716	5	+0.055
<i>K. pneumoniae</i> –carbapenems	5	+0.710	5	+0.219
<i>S. pneumoniae</i> –macrolides	1	+0.548	1	+0.141
<i>E. coli</i> –3rd-gen cephalosporins	5	+0.453	3	–0.174
<i>S. aureus</i> –MRSA	— (no class-specific driver)	—	3	+0.117

Note. Pooled correlations treat all country–year observations together and are computed on the class-relevant consumption series; *S. aureus*–MRSA has no such series and is reported for the within-country analysis only, using total consumption. Within-country correlations are computed after subtracting each country’s own mean from both variables, which removes persistent between-country differences and isolates year-to-year temporal covariation.

RQ2 — Forecast Accuracy

Covariate-based models explain meaningful variance in resistance on years never seen during training, with held-out R^2 between 0.670 and 0.733 depending on the learner (Tables 13 and 14). These positive hold-out values provide empirical evidence against the null benchmark of non-positive predictive power. No confidence interval or formal significance test was computed for R^2 , so RQ2 is answered on the strength of the observed out-of-sample performance rather than by a formal inferential rejection. Forecast accuracy is highest for combinations with strong, long lags—

E. coli–fluoroquinolones and *K. pneumoniae*–carbapenems—and weakest for MRSA, which lacks a clean single-class consumption driver.

RQ3 — Driver Importance

Permutation importance on held-out data identifies population density and lagged consumption as the dominant predictors, with the two-year and five-year lags outweighing current-year consumption (Table 17). This independently reinforces the RQ1 finding that the consumption effect is delayed. It should be noted that permutation importance yields a ranking rather than a formal significance test: the results provide evidence that predictor contributions are not uniform, but the RQ3 null hypothesis of equal contribution is not rejected by a formal inferential procedure. A bootstrap or permutation-distribution test of the importance differences would be required for a strict hypothesis decision, and is identified as future work. One result warrants explicit comment: population density shows almost no linear correlation with resistance ($r = 0.10$ in Figure 6) yet ranks as the strongest single predictor in the fitted model. This is not a contradiction but a demonstration that density operates non-linearly and through interactions that a correlation coefficient cannot detect—precisely the justification for selecting a tree-based learner over a purely linear specification.

Table 17

RQ3 — gradient-boosted feature importance

Feature	Importance	Note
pop_density	0.357	Strongest single predictor; non-linear effect
ddd_lag2	0.168	Consumption two years earlier
ddd_lag5	0.141	Consumption five years earlier
ddd_class_t	0.130	Current-year consumption
ddd_lag3	0.098	Consumption three years earlier
ddd_lag4	0.065	Consumption four years earlier
ddd_lag1	0.041	Consumption one year earlier

Note. Climate (degree days) contributed comparatively little at this stage, partly because the indicator is available only for 2016–2024; extending its coverage is identified as future work.

RQ4 — Pooled Model versus Baselines

The pooled cross-country models outperform the naive baseline decisively on directional accuracy (0.497–0.546 versus 0.000), though as Table 15 shows the naive baseline remains significantly

ahead on absolute error while losing on raw error and R^2 , as Table 12 shows. The RQ4 null hypothesis is rejected on the directional criterion, which is the operationally meaningful one for early warning. It is worth stating plainly that naive persistence remains difficult to beat on year-over-year error: because resistance changes slowly, predicting no change is a strong error-minimising strategy that nonetheless carries no forecasting value. Recognising this distinction, rather than reporting R^2 alone, is central to interpreting these results correctly.

Overall Interpretation

Taken together, the results indicate that a lag-aware, covariate-enriched model captures the direction of change in resistance that a naive persistence model cannot, even though persistence remains hard to beat on raw error. The dominance of lagged consumption and demographic context confirms that both prescribing history and population structure shape resistance trajectories. Performance varies systematically by pathogen: combinations with strong, long lags are the most predictable, while MRSA is least amenable to consumption-based forecasting. The overfitting analysis further indicates that the models are flexible relative to the available data, so reported test figures should be treated as the realistic performance envelope.

Practical Significance

For a stewardship analyst, the value of the pipeline lies in its directional early-warning signal. A model that indicates, for the coming year, that fluoroquinolone or carbapenem resistance is likely to rise in a given country allows prescribing guidelines and infection-control resources to be adjusted before resistance becomes entrenched. The operational value of the long lags is different but complementary: because consumption from up to five years earlier still carries predictive weight, a prescribing change made now continues to influence the forecast for several years, which tells stewardship committees how durable a policy change is likely to be rather than how far ahead the model can see.

Implementation and User Benefit

Deployment Approach

The pipeline can be packaged as a lightweight scheduled job that re-runs whenever ECDC publishes updated EARS-Net and ESAC-Net figures—in practice, an annual cycle. Because it depends only on open data and CPU-only Python libraries, it can be deployed as a container on any cloud platform or served as a small REST API behind a dashboard, with no specialised hardware.

System Integration

The pipeline consumes the same DDD and resistance exports that national surveillance teams already download, so it integrates into existing ECDC and national workflows without imposing any new data-collection burden. Outputs—per-combination forecasts and driver rankings—can be surfaced inside existing surveillance dashboards.

User Interaction

A stewardship analyst selects a country and a pathogen–antibiotic combination and receives the forecast resistance trajectory, the predicted direction of change, and the ranked drivers behind it. No coding is required at the point of use.

Benefits to Users

Operationally, the tool prioritises where limited stewardship attention should be directed. Strategically, its interpretable driver rankings support the evidence case for specific prescribing interventions. Financially, reliance on open data and CPU-only computation keeps running costs negligible; the dominant operational risk is not cost but data continuity—a change to ECDC export formats or to the ATC/DDD index (as occurred in the 2019 revision) would require the cleaning stage to be updated, which is mitigated by a schema-validation step at ingestion.

Example Use Case

An analyst in a southern-European country observes rising fluoroquinolone consumption in the latest ESAC-Net release. The pipeline, trained on the pan-European panel, produces a one-year-

ahead forecast indicating that *E. coli*–fluoroquinolone resistance is likely to continue rising, and identifies lagged quinolone consumption as a dominant contributor to that prediction. Because consumption from up to five years earlier still carries weight, the elevated forecast is unlikely to reverse quickly. The forecast provides additional evidence for the stewardship committee when considering whether targeted reductions in outpatient fluoroquinolone prescribing are warranted; it does not by itself establish that reducing prescribing will cause resistance to fall, since the underlying design is observational.

Limitations and Further Improvements

Limitations

- Consumption is reported by ATC class while resistance is drug-specific; carbapenems and cephalosporins share group J01D, limiting granularity for those pairs.
- *S. aureus*–MRSA lacks a clean single-class consumption driver and is excluded from the consumption-based lag models.
- Climate indicators (degree days) cover only 2016–2024, so the climate contribution in RQ3 is underpowered relative to consumption and demographics.
- Several countries have sparse consumption reporting; the panel is unbalanced and these countries contribute fewer modelling rows.
- High autocorrelation inflates R^2 for naive prediction, which is why directional accuracy is emphasised alongside error metrics.
- Surveillance methods changed over time, notably the 2019 ATC/DDD index revision, a comparability caveat documented by ECDC (2025).
- The panel structure means repeated yearly measurements per country are not independent, so classical p-values are approximate; a mixed-effects specification would be the fully rigorous inferential route.

- The design is observational, so the findings establish association consistent with a known biological mechanism rather than proof of causation.
- Short interior gaps in consumption were filled by linear interpolation within each country–combination series before the temporal split. Because interpolation between two known points uses a later value to inform an earlier one, it introduces a limited risk of look-ahead leakage for the interpolated cells. The affected cells are a small fraction of the panel and the target variable was never interpolated, but a fully leakage-proof design would restrict interpolation to the training window and forward-fill within the test window.
- Much of the raw consumption–resistance correlation is cross-country rather than temporal. When country means are removed, the within-country partial correlations fall to between -0.17 and $+0.22$, indicating that a substantial part of the pooled association reflects persistent differences between national health systems rather than year-to-year dynamics within a country. Pooled ecological correlations should therefore be interpreted with caution.

Impact of Limitations

These limitations principally affect the precision of driver attribution and the generalisability of the climate findings rather than the central lag result, which is robust and consistent in direction across all combinations tested. The J01D granularity constraint may understate class-specific effects for the carbapenem and cephalosporin pairs, and the unbalanced panel means country-level forecasts are less reliable for states with sparse consumption histories. The non-independence of panel observations means reported significance levels should be read as approximate rather than exact. The gap between pooled and within-country correlations further implies that the lag structure identified in RQ1 is best read as a population-level regularity rather than a within-country causal dose–response relationship.

Future Improvements

- Apply stronger regularisation to the gradient-boosted model to close the generalisation gap identified in Table 14.
- Extend or substitute the climate covariate to cover the full 2000–2024 period, giving the climate hypothesis adequate power.
- Fit a mixed-effects model with country as a random effect to obtain rigorous inference under the panel structure.
- Benchmark against dedicated time-series specifications such as ARIMA or SARIMAX fitted per country–combination. These were not evaluated here because the panel offers only about 22 annual observations per series — short for reliable ARIMA identification — and because univariate specifications cannot exploit the cross-country pooling that is central to this design; nonetheless they remain the standard forecasting comparator and would strengthen the RQ4 benchmark.
- Add SHAP-based per-pathogen driver analysis (Lundberg & Lee, 2017) for finer-grained attribution than permutation importance provides.

Future Scope

The framework generalises beyond the five combinations studied here to any pathogen–antibiotic pair with adequate surveillance coverage. It could be extended with additional covariates—hospital bed density, international travel volume, or veterinary antibiotic use—to build a comprehensive One-Health resistance forecasting system, and the same lag-aware temporal-validation design would transfer directly to other settings where an intervention’s effect is delayed and must be anticipated rather than observed.

References

Aldeyab, M. A., Monnet, D. L., López-Lozano, J. M., Hughes, C. M., Scott, M. G., Kearney, M. P., Lawson, D. H., & McElnay, J. C. (2008). Modelling the impact of antibiotic use and

- infection control practices on the incidence of hospital-acquired MRSA: A time-series analysis. *Journal of Antimicrobial Chemotherapy*, 62(3), 593–600.
- Bell, B. G., Schellevis, F., Stobberingh, E., Goossens, H., & Pringle, M. (2014). A systematic review and meta-analysis of the effects of antibiotic consumption on antibiotic resistance. *BMC Infectious Diseases*, 14, 13. <https://doi.org/10.1186/1471-2334-14-13>
- Chatterjee, A., Modarai, M., Naylor, N. R., Boyd, S. E., Atun, R., Barlow, J., Holmes, A. H., Johnson, A., & Robotham, J. V. (2018). Quantifying drivers of antibiotic resistance in humans: A systematic review. *The Lancet Infectious Diseases*, 18(12), e368–e378. [https://doi.org/10.1016/S1473-3099\(18\)30296-2](https://doi.org/10.1016/S1473-3099(18)30296-2)
- Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 785–794). Association for Computing Machinery.
- European Centre for Disease Prevention and Control. (2025). *Antimicrobial consumption in the EU/EEA (ESAC-Net): Annual epidemiological report for 2024*. ECDC.
- Goossens, H., Ferech, M., Vander Stichele, R., & Elseviers, M. (2005). Outpatient antibiotic use in Europe and association with resistance: A cross-national database study. *The Lancet*, 365(9459), 579–587. [https://doi.org/10.1016/S0140-6736\(05\)17907-0](https://doi.org/10.1016/S0140-6736(05)17907-0)
- Kirby, A., Herbert, A., & Sheridan, E. (2022). Estimating the consumption–resistance relationship for antibiotics using European surveillance data. *Journal of Global Antimicrobial Resistance*, 30, 428–434.
- Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems 30* (pp. 4765–4774).
- MacFadden, D. R., McGough, S. F., Fisman, D., Santillana, M., & Brownstein, J. S. (2018). Antibiotic resistance increases with local temperature. *Nature Climate Change*, 8(6), 510–514. <https://doi.org/10.1038/s41558-018-0161-6>
- Murray, C. J. L., Ikuta, K. S., Sharara, F., Swetschinski, L., Robles Aguilar, G., Gray, A., Han, C., Bisignano, C., Rao, P., Wool, E., & Naghavi, M. (2022). Global burden of bacterial

antimicrobial resistance in 2019: A systematic analysis. *The Lancet*, 399(10325), 629–655. [https://doi.org/10.1016/S0140-6736\(21\)02724-0](https://doi.org/10.1016/S0140-6736(21)02724-0)

Olesen, S. W., Barnett, M. L., MacFadden, D. R., Brownstein, J. S., Hernandez-Diaz, S., Lipsitch, M., & Grad, Y. H. (2018). The distribution of antibiotic use and its association with antibiotic resistance. *eLife*, 7, e39435.

Appendix A: Data Dictionary and Panel Construction

Appendix A documents the variable set (Table 2 in the main text) and the construction of the merged panel. The panel is assembled by `data_prep.py`, which loads each of the four sources, harmonises country identifiers to ISO-2 codes, pivots ESAC-Net consumption to one column per ATC class, reshapes the Eurostat degree-day and population-density spreadsheets from wide to long format, and left-joins all sources onto the EARS-Net resistance outcome by country and year. The resulting file, `amr_panel.csv`, contains 3,257 rows and 18 columns and is the single input to all downstream analysis reported in this document.

Appendix B: Analysis Code

The complete Python code reproducing every figure and table in this report is available in the project GitHub repository listed after the title page. The repository is organised as three scripts referenced throughout the text: `data_prep.py` (panel construction, described in Appendix A); `01_eda.py` (the univariate, bivariate, and temporal analyses underlying Figures 2 through 6 and Tables 11 and 12); and `02_modeling.py` (the three-way temporal split, expanding-window cross-validation, and the RQ1–RQ4 analyses reported in Tables 13 through 17 and Figure 8). See Appendix B in the repository for the code used to generate Figure 5 and the correlation analysis discussed in the Exploratory Data Analysis section.